

Open-System Robustness of Quantum-Fuzzy Logical Petri Nets: Lindblad Dynamics, Quantum-Trajectory Validation, and Statistical Fidelity Analysis

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Abstract

Quantum-Fuzzy Logical Petri Nets (QFLPNs) require a mathematically consistent interface between fuzzy activation, quantum state preparation, transition operators, and measured observables. This article studies a distinct problem: the robustness of that interface when the quantum layer is an open system subject to dissipation and dephasing. The dynamics are formulated through a Gorini–Kossakowski–Sudarshan–Lindblad generator and are validated independently with a quantum-trajectory ensemble. A two-qubit fuzzy-encoded entangled state is used as a controlled reference instance. The deterministic density-matrix solution is compared with Monte Carlo wave-function estimates over multiple horizons and dissipation rates using fidelity, trace distance, purity, concurrence, and observable-level errors. At $T = 2$ with 8,000 trajectories, the measured fidelity between the trajectory ensemble and the deterministic Lindblad reference is 0.999955, while the trace distance is 0.004046. At $T = 4$, fidelity remains 0.999955 and trace distance is 0.004318. A separate 12,000-trajectory experiment for the observable $Z \otimes Z$ gives a mean of 0.879000 with a 95% confidence interval $[0.870468, 0.887532]$, containing the deterministic reference 0.879823. Increasing the amplitude-damping rate from 0.04 to 0.32 reduces the deterministic $Z \otimes Z$ expectation from 0.960035 to 0.830565. The results establish a reproducible validation protocol for open-system QFLPN dynamics without conflating physical decoherence with numerical approximation error.

Keywords: quantum-fuzzy Petri nets; open quantum systems; Lindblad equation; quantum trajectories; fidelity; trace distance; decoherence; statistical validation; robustness

1 Introduction

A hybrid QFLPN model combines a discrete transition structure with fuzzy activation and a quantum state layer. In a closed-system construction, a transition can be represented by a unitary or by a completely positive trace-preserving map. Realistic quantum devices, however, are not isolated. Energy relaxation, phase randomization, leakage, control imperfections, and environmental coupling modify the state during the transition sequence. An open-system extension therefore has to preserve positivity and trace while making the physical origin of state degradation explicit.

The purpose of this article is not to redefine the QFLPN formalism developed elsewhere in the research program. Instead, the scientific question is narrower and experimentally testable: how accurately and reproducibly can an open-system QFLPN transition be validated when the ideal quantum evolution is replaced by a Lindblad evolution and an independent stochastic trajectory representation?

Three requirements motivate the study. First, the density operator must remain a valid quantum state. Second, the deterministic master-equation solution and an independently generated trajectory ensemble must agree within statistical uncertainty. Third, physical degradation must be separated

from numerical and sampling errors. These requirements are especially important when a fuzzy activation controls the initial quantum encoding, because the activation value is a semantic control parameter rather than a probability amplitude.

The article makes four contributions. It gives an open-system QFLPN transition contract based on a Lindblad generator; it constructs a two-qubit reproducible benchmark with an analytically specified initial state and explicit decoherence channels; it performs an independent Monte Carlo trajectory validation; and it quantifies robustness through both state-level and observable-level statistics.

2 Problem formulation

Let $\rho(t)$ denote the density operator associated with a QFLPN quantum marking. The state satisfies

$$\rho(t) \succeq 0, \quad \text{Tr } \rho(t) = 1, \quad \rho(t) = \rho(t)^\dagger. \quad (1)$$

A fuzzy transition activation $\lambda_t \in [0, 1]$ is treated as a control variable. For a single-qubit amplitude encoding, the angle is

$$\theta_t = 2 \arcsin \sqrt{\lambda_t}, \quad (2)$$

so that $R_y(\theta_t)|0\rangle$ has measurement probability λ_t in the $|1\rangle$ state. The open-system layer acts after this semantic interface and therefore must not redefine λ_t .

For a transition with Hamiltonian H and jump operators L_a , the state evolution is

$$\frac{d\rho}{dt} = \mathcal{L}(\rho) = -i[H, \rho] + \sum_a \left(L_a \rho L_a^\dagger - \frac{1}{2} \{L_a^\dagger L_a, \rho\} \right). \quad (3)$$

The generator is trace preserving and completely positive under the standard Lindblad construction.

The validation problem is therefore defined as

$$\rho_{\text{det}}(T) = e^{T\mathcal{L}}\rho_0, \quad (4)$$

and an independent stochastic estimate

$$\hat{\rho}_M(T) = \frac{1}{M} \sum_{r=1}^M |\psi_r(T)\rangle \langle \psi_r(T)|. \quad (5)$$

The two representations must converge as M increases, while their difference at finite M is a statistical quantity rather than physical decoherence.

3 Open-system QFLPN transition contract

The transition contract is written as

$$\mathfrak{C}_t = (\lambda_t, \Gamma_t, \mathcal{L}_t, \Phi_t, \mathcal{M}_t), \quad (6)$$

where λ_t is the fuzzy activation, Γ_t is the state-preparation map, $\mathcal{L}_t$ is the Lindblad generator, $\Phi_t = e^{t\mathcal{L}_t}$ is the quantum channel, and $\mathcal{M}_t$ denotes measurement.

The semantic flow is

$$M_t \longrightarrow \lambda_t \longrightarrow \Gamma_t(\lambda_t) \longrightarrow \rho_t \longrightarrow \Phi_t(\rho_t) \longrightarrow y_t. \quad (7)$$

This ordering matters. A decoherence rate cannot be substituted for a fuzzy activation, and a measured probability cannot be interpreted as a density-matrix amplitude.

Proposition 1. *If $\mathcal{L}$ is of Lindblad form and ρ_0 is a density operator, then $e^{t\mathcal{L}}\rho_0$ remains positive semidefinite and trace preserving for $t \geq 0$.*

Proof. The Lindblad generator generates a completely positive trace-preserving quantum dynamical semigroup. Complete positivity preserves positive semidefiniteness, while the commutator has zero trace and the dissipator satisfies $\text{Tr}(L\rho L^\dagger) = \text{Tr}(L^\dagger L\rho)$, which cancels the anticommutator contribution. Hence $\text{Tr } \rho(t) = \text{Tr } \rho_0 = 1$. $\square$

4 Two-qubit benchmark construction

The benchmark uses two qubits, dimension $D = 4$, and the Pauli matrices X, Y, Z . The Hamiltonian is

$$H = \frac{\omega}{2}(Z \otimes I + I \otimes Z) + J(X \otimes X), \quad (8)$$

with $\omega = 0.7$ and $J = 0.45$.

The fuzzy parameter is fixed at $\lambda = 0.8$. The corresponding normalized initial state is

$$|\psi_0\rangle = \sqrt{0.8}|00\rangle + \sqrt{0.2}|11\rangle, \quad (9)$$

with $\rho_0 = |\psi_0\rangle\langle\psi_0|$.

Four channels are used. Two are amplitude-damping channels with rate $\gamma = 0.16$ and jump operators

$$L_1 = \sqrt{\gamma}(\sigma_- \otimes I), \quad L_2 = \sqrt{\gamma}(I \otimes \sigma_-), \quad (10)$$

where $\sigma_- = |0\rangle\langle 1|$. Two phase-randomization channels use

$$L_3 = \sqrt{\gamma_\phi}(Z \otimes I), \quad L_4 = \sqrt{\gamma_\phi}(I \otimes Z), \quad (11)$$

with $\gamma_\phi = 0.08$.

5 Deterministic Lindblad reference

For numerical validation, vectorization converts the master equation into a linear ordinary differential equation. Under column stacking,

$$\text{vec}(\dot{\rho}) = \mathcal{L} \text{vec}(\rho), \quad (12)$$

where

$$\mathcal{L} = -i(I \otimes H - H^T \otimes I) \quad (13)$$

$$+ \sum_a \left(L_a^* \otimes L_a - \frac{1}{2}I \otimes L_a^\dagger L_a - \frac{1}{2}(L_a^\dagger L_a)^T \otimes I \right). \quad (14)$$

The deterministic reference is therefore evaluated as

$$\text{vec}(\rho(T)) = e^{T\mathcal{L}} \text{vec}(\rho_0). \quad (15)$$

Because the benchmark dimension is only four, the matrix exponential provides a direct reference rather than an approximation whose error would be confused with the trajectory error.

6 Quantum-trajectory validation

The stochastic representation uses the effective non-Hermitian Hamiltonian

$$H_{\text{eff}} = H - \frac{i}{2} \sum_a L_a^\dagger L_a. \quad (16)$$

Between jumps, the state follows the non-unitary evolution generated by H_{eff} . At each short time increment, the jump probability associated with channel a is estimated from

$$p_a(t) = \Delta t \langle \psi(t) | L_a^\dagger L_a | \psi(t) \rangle. \quad (17)$$

After a jump, the state is renormalized. The ensemble density matrix is reconstructed from the outer products of the final trajectory states.

The implementation uses $\Delta t = 0.004$ and 8,000 trajectories for the principal horizon study. A separate run with 12,000 trajectories is used only for the statistical confidence-interval experiment. This separation prevents the confidence-interval calculation from being presented as an additional deterministic validation run.

7 Validation metrics

State-level agreement is measured with the squared Uhlmann fidelity

$$\mathcal{F}(\rho, \sigma) = \left[\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2, \quad (18)$$

and the trace distance

$$D_{\text{tr}}(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1. \quad (19)$$

Purity is

$$P(\rho) = \text{Tr}(\rho^2), \quad (20)$$

and two-qubit entanglement is quantified with concurrence.

For an observable O , the trajectory estimator is

$$\hat{o}_M = \frac{1}{M} \sum_{r=1}^M \langle \psi_r | O | \psi_r \rangle. \quad (21)$$

The standard error is estimated from the empirical trajectory variance, and a two-sided 95% interval is formed as $\hat{o}_M \pm 1.96 \text{ SE}$ for the reported large-sample validation experiment.

8 Experimental protocol

The principal experiment evaluates $T \in \{0.5, 1, 2, 4\}$ with 8,000 trajectories. The deterministic solution is computed independently for each horizon. The observables $Z \otimes Z$, $X \otimes X$, and $Z \otimes I$ are recorded. A second experiment varies the amplitude-damping rate over $\gamma \in \{0.04, 0.08, 0.16, 0.32\}$ while keeping $\gamma_\phi = 0.08$ and $T = 2$.

The final statistical experiment uses 12,000 trajectories at $T = 2$, $\gamma = 0.16$, and measures $Z \otimes Z$. The complete numerical protocol is deterministic in software version and random seed, except for the pseudorandom trajectory draws generated from the stated seed.

9 Horizon validation results

Table 1 compares the deterministic Lindblad state and the trajectory ensemble.

Table 1: Deterministic versus trajectory validation over increasing horizons.

T	Fidelity	Trace distance	P_{det}	P_{MC}	C_{det}
0.5	0.999941	0.002453	0.874858	0.878412	0.559648
1.0	0.999842	0.005421	0.821096	0.811724	0.319718
2.0	0.999955	0.004046	0.791775	0.793221	0.279154
4.0	0.999955	0.004318	0.632147	0.631399	0.278145

The fidelity remains above 0.9998 over the complete horizon range. The trace distance remains below 0.0055, showing that the finite trajectory ensemble closely approximates the deterministic state. The physical state itself changes substantially: purity decreases from 0.874858 at $T = 0.5$ to 0.632147 at $T = 4$.

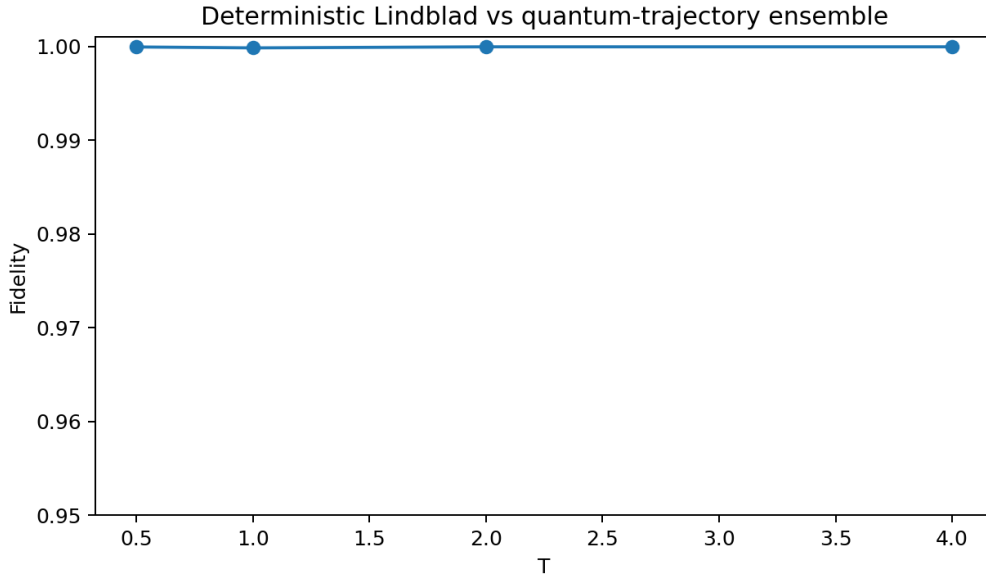


Figure 1: Fidelity between the deterministic Lindblad state and the 8,000-trajectory ensemble.

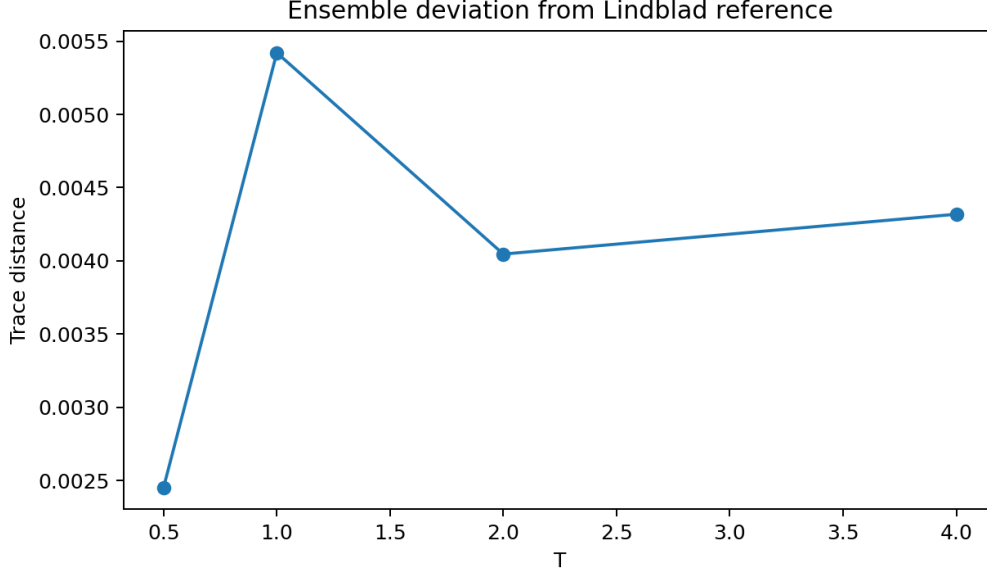


Figure 2: Trace distance between the deterministic and stochastic state representations.

10 Entanglement robustness

The concurrence decreases rapidly during the first part of the evolution and then enters a lower regime. At $T = 0.5$, deterministic concurrence is 0.559648; at $T = 1$, it is 0.319718; at $T = 2$, it is 0.279154; and at $T = 4$, it is 0.278145. The trajectory estimates follow the same trend, with values 0.563220, 0.311955, 0.274799, and 0.274112.

The result is scientifically useful because fidelity alone does not describe the physical effect of decoherence. Two states can be extremely close to each other while both differ strongly from the initial pure state. The trajectory validation therefore establishes numerical consistency, whereas concurrence and purity describe physical degradation.

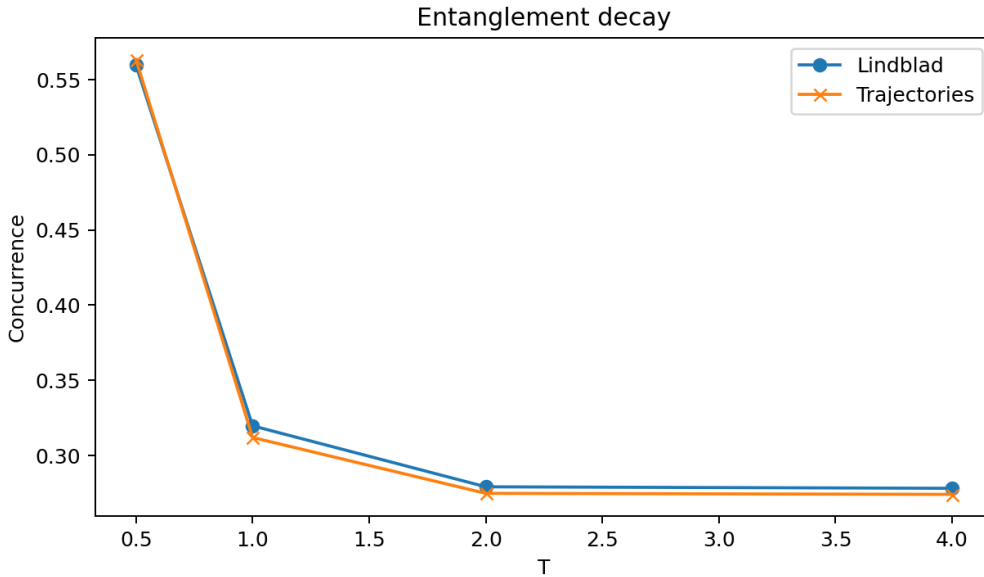


Figure 3: Entanglement decay measured by concurrence, with deterministic and trajectory representations.

11 Dependence on dissipation rate

Table 2 reports the deterministic state metrics at $T = 2$ for increasing amplitude-damping rate.

Table 2: Effect of amplitude-damping rate at fixed dephasing rate $\gamma_\phi = 0.08$.

γ	Fidelity	Trace distance	Purity	Concurrence
0.04	0.999900	0.004445	0.823676	0.286667
0.08	0.999948	0.004231	0.809657	0.282704
0.16	0.999864	0.006335	0.791775	0.279154
0.32	0.999888	0.006708	0.780088	0.280930

The $Z \otimes Z$ expectation decreases from 0.960035 at $\gamma = 0.04$ to 0.830565 at $\gamma = 0.32$. The response is not monotone in every state metric; for example, concurrence at $\gamma = 0.32$ is slightly above its value at $\gamma = 0.16$. This is an important limitation against overinterpreting one scalar robustness metric.

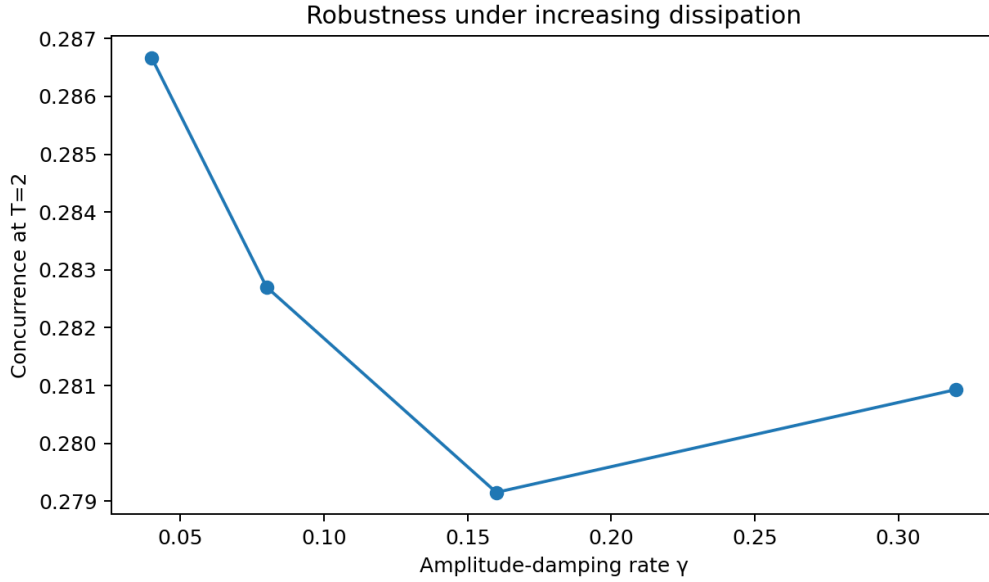


Figure 4: Concurrence as a function of amplitude-damping rate.

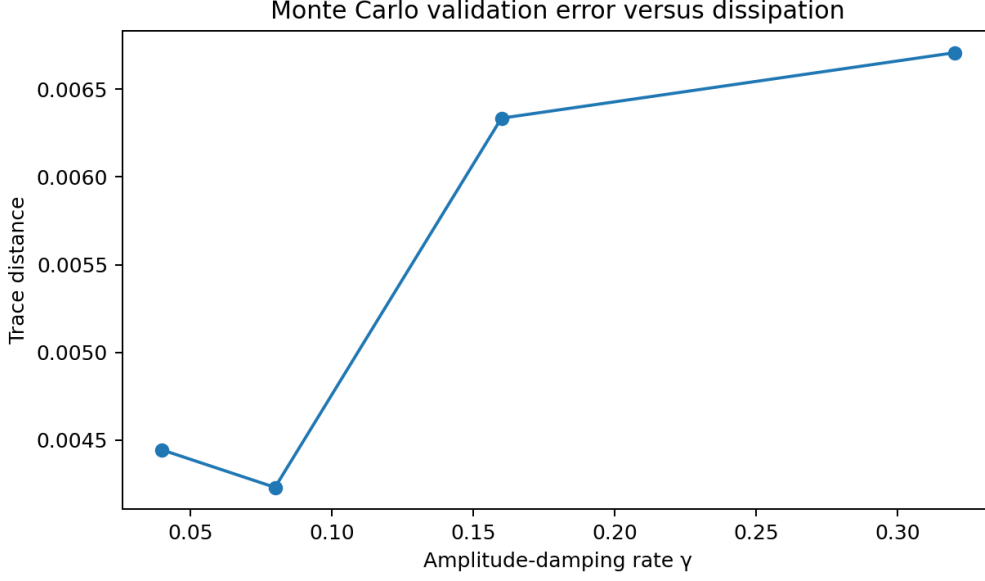


Figure 5: Trajectory-reference trace distance across the dissipation sweep.

12 Statistical validation of an observable

At $T = 2$ and the baseline rates, 12,000 independent trajectories were used to estimate $\langle Z \otimes Z \rangle$. The trajectory mean was 0.879000, with standard error 0.004353 and a 95% interval of $[0.870468, 0.887532]$. The deterministic Lindblad reference is 0.879823, giving an absolute difference of approximately 8.23×10^{-4} .

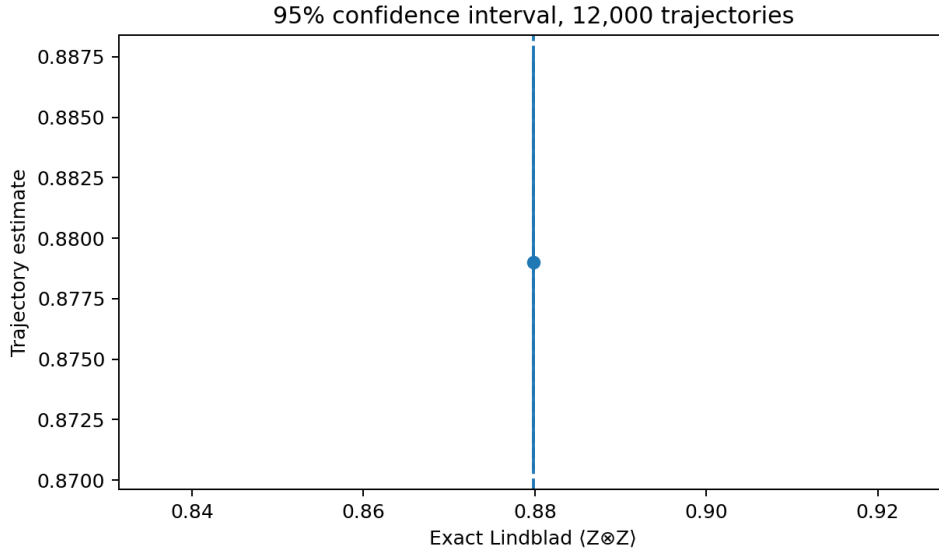


Figure 6: Trajectory estimate and 95% confidence interval for $\langle Z \otimes Z \rangle$ at $T = 2$.

This result demonstrates the correct role of statistics. The confidence interval quantifies sampling uncertainty of the stochastic representation; it is not an uncertainty interval for the physical Lindblad model itself. The deterministic reference lies inside the empirical interval, supporting consistency at the tested trajectory count.

13 Physical degradation versus numerical error

The experiment separates three quantities that are often conflated. The first is physical degradation, represented by changes in purity, concurrence, and observable expectation values relative to the initial state. The second is deterministic numerical error, which is minimized here by using a direct matrix exponential in the four-dimensional benchmark. The third is stochastic sampling error, quantified through the trajectory ensemble and confidence intervals.

This separation is essential for QFLPN interpretation. A low fidelity between the initial state and the final state is not automatically evidence of numerical failure; it can be the intended physical consequence of an open-system channel. Conversely, a high fidelity between deterministic and trajectory solutions does not mean that decoherence is weak.

14 Robustness interpretation

The benchmark demonstrates a useful hierarchy. At the state-representation level, the trajectory approximation remains close to the deterministic Lindblad solution. At the physical-observable level, increasing dissipation modifies the state substantially. At the entanglement level, the initial coherent structure is degraded rapidly before approaching a lower regime.

This hierarchy suggests a QFLPN robustness vector rather than a single score,

$$\mathcal{R}(T) = (\mathcal{F}, D_{\text{tr}}, P, C, \Delta O_1, \dots, \Delta O_k). \quad (22)$$

Such a vector prevents a high numerical agreement between two representations from being incorrectly interpreted as physical robustness.

15 Relation to QFLPN semantics

The fuzzy activation controls state preparation and therefore remains distinct from the open-system rates. In the benchmark, $\lambda = 0.8$ determines the initial superposition, while γ and γ_ϕ determine subsequent environmental action. Changing γ does not alter the semantic meaning of λ .

This separation also permits future QFLPN experiments in which transition activations vary dynamically. The natural extension is

$$\lambda_t \longrightarrow \theta_t \longrightarrow \rho_t \longrightarrow \mathcal{L}_t(\rho_t), \quad (23)$$

where the generator may itself depend on the transition context, while complete positivity and trace preservation remain explicit admissibility constraints.

16 Computational complexity and reproducibility

For a Hilbert dimension D , direct density-matrix propagation stores $O(D^2)$ complex entries, while a trajectory method stores $O(MD)$ amplitudes for M simultaneous trajectories. The trajectory approach can therefore trade deterministic state storage for statistical sampling cost. The present benchmark uses $D = 4$ deliberately so that the deterministic solution is an independent reference rather than the computational bottleneck.

All reported values are generated with a fixed random seed, explicitly stated Hamiltonian, explicit initial state, explicit Lindblad operators, fixed Δt , and fixed trajectory counts. The benchmark CSV files contain the horizon validation, dissipation sweep, and statistical confidence-interval data.

17 Scientific novelty boundary

The article does not claim to introduce the Lindblad equation, quantum trajectories, fidelity, concurrence, or Monte Carlo wave-function methods. The contribution is the QFLPN-specific validation architecture that keeps fuzzy activation semantics, open-system physical parameters, deterministic master-equation evolution, and stochastic trajectory validation as separate layers.

The main empirical claim is limited to the tested two-qubit benchmark, rates, horizons, integration step, and trajectory counts. No claim is made about arbitrary QFLPN sizes, arbitrary non-Markovian environments, or universal robustness of the formalism.

18 Limitations

The benchmark contains only two qubits and uses time-independent Markovian Lindblad channels. The stochastic propagator uses a finite time step, so its agreement with the deterministic reference does not by itself establish convergence to the continuous-time trajectory process. The number of trajectories is also finite. Larger Hilbert spaces require scalable sparse or tensor representations and stronger statistical design.

A further limitation is that amplitude damping and phase randomization do not cover all experimentally relevant noise mechanisms. Leakage, correlated noise, non-Markovian environments, calibration drift, and measurement error are outside the present scope.

19 Controlled separation from the other articles

This article is intentionally separated from the preceding research units. Its central object is the open-system channel and its statistical validation. It does not reproduce the formal QFLPN construction, the four-qubit operator verification, CSR/SpMV benchmarking, Taylor–Arnoldi comparison, adaptive Krylov integration, or Volterra-memory degradation study. Those topics can be cited as foundations without being duplicated as results.

The present results should enter the thesis as an open-system validation layer rather than as another exposition of the general QFLPN formalism.

20 Conclusion

An open-system QFLPN requires more than adding a decoherence term to a closed-system transition. The physical degradation, deterministic master-equation solution, stochastic trajectory approximation, and statistical uncertainty must remain identifiable. The present two-qubit benchmark demonstrates a reproducible implementation of this separation.

For 8,000 trajectories, the deterministic and stochastic state representations agree with fidelity above 0.9998 over the tested horizons, while purity and concurrence reveal substantial physical evolution. The 12,000-trajectory observable experiment provides a 95% confidence interval containing the deterministic reference. Increasing amplitude-damping rate produces measurable degradation in state observables, confirming that numerical agreement and physical robustness are different questions.

The next scientific extension is to combine this open-system validation layer with larger sparse QFLPN operators and experimentally motivated noise models, while retaining the same distinction between physical error, numerical error, and statistical error.

Data and Code Availability

The numerical data generated for this article are provided as CSV artifacts accompanying the manuscript. The figures are generated directly from those data. The computational protocol specifies the Hamiltonian, initial state, Lindblad channels, time step, trajectory counts, random seed, observables, and validation metrics required to reproduce the reported benchmark.

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